

Forecasting the Onset of Hallucination: Causal, Token-Level, Black-Box Survival Analysis During Generation

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Abstract

Token-level hallucination detectors react only after a hallucination has begun; a companion analysis proves a change-point delay bound of about 1.3 tokens after the onset. We ask whether the onset can instead be *forecast*, before it begins. We cast the task as discrete-time survival analysis: a per-token hazard over the risk set of still-faithful tokens, computed from causal, streaming-safe, black-box features and read by a forward-only recurrent head. On the RAGTruth corpus the onset is forecastable at every horizon, reaching 0.777 ± 0.002 pooled area under the ROC curve (AUROC) at horizon $k=3$ (onset within three tokens) over five seeds. A pooled score, however, also rewards knowing which document is risky; isolating timing within a single document, where any prompt-level predictor is uninformative, the forecaster still ranks imminent-onset tokens at 0.687 AUROC, the genuine timing signal no prompt-level predictor supplies. The early warning is text-novelty drift, not language-model surprisal, and it is neither onset-autocorrelation nor self-excitation. At a matched false-alarm budget it warns a median of 11 tokens before onset where the detector fires 15 after, exceeding by $6.4\times$ a ceiling we prove for any monitor whose still-faithful prefix is uninformative about the onset. The signal is uneven: it is strongest where an output is tightly bound to its source and near chance on summarization, and it does not transfer zero-shot. A detector can only react once a hallucination has begun; a forecaster warns while the text is still faithful. Code: <https://github.com/YehudaItkin/hallucination-onset-forecasting>.

1 Introduction

A hallucination detector that flags a false claim after the model has written it has already lost the reader. Token-level detectors are trained as classifiers. Even a streaming monitor built on quickest change detection reacts only after the change. Our companion analysis proves a delay bound of about 1.3 tokens after onset at a 1% false-alarm rate, and the learned detectors that approach it still fire 11 to 13 tokens late among the onsets they catch [11]. Detection, by construction, comes after the event. A monitor that only reacts can log or redact after the fact. A monitor that is to *act*, to abstain or re-decode or steer, needs warning before the first false token.

This is the third question in a sequence. The first asked *what* and *where*: a temporal multi-signal model localizes hallucinated spans at the token level from a 33-dimensional black-box feature stream of text-novelty, entailment, and language-model signals [12]. The second asked *how fast* one can react once a span begins, and answered with a change-point delay bound [11]. Here we ask whether the onset can be *forecast*. Given only what a streaming decoder has emitted up to token t , can we predict that a hallucination will begin within the next few tokens, before it does?

The onset is a promising place to predict. The signal concentrates at the first hallucinated token [37], and a hallucinated state propagates forward through autoregressive conditioning [1], so

the state just before an onset should already differ from a stably faithful one. What is missing is a predictor that reads that difference *causally*, during generation, and points forward in time. A parallel line does estimate hallucination risk before decoding, but at the prompt or response level and from model internals [24, 31, 17]: it asks *whether* a generation is risky. We ask *when and where* inside one the risk becomes an onset (Section 2). Prompt-level risk chooses which generations to watch; we say when to intervene inside them.

We formulate forecasting as discrete-time survival analysis. We treat the onset of a hallucinated span as a failure event over the risk set of still-faithful tokens, and model a per-token hazard [7, 36, 38]: the probability that an onset falls within the next k tokens, given the causal feature history. Two properties make the predictor deployable. It uses only *causal* features: we audit the 33-dimensional stream and drop every component that peeks at the full output, leaving 24 streaming-safe features. It also reads them with a *forward-only* recurrent network, so the hazard at token t depends on nothing after t . We ask three questions.

- RQ1** Can the onset be forecast from causal, black-box, token-level features before it begins, and how far ahead? Does the forecast measure timing inside a generation, not just which generation is risky?
- RQ2** Where does the early warning live: in the model’s uncertainty, or in the text it produces, and is it a real content signal rather than autocorrelation of the onset process?
- RQ3** Does forecasting buy anything a detector cannot? Does it reach the negative delay a change-point bound forbids, and does it transfer across domains?

For RQ1, the hazard is forecastable at every horizon. A recurrent hazard head reaches 0.777 ± 0.002 AUROC at horizon $k=3$ (onset within the next three tokens; five seeds), 12 points above a pointwise logistic hazard, and the causal features lose almost nothing to the future-leaking ceiling. A pooled AUROC, however, blends two different things: whether a document hallucinates at all, and when inside it an onset is coming. An oracle simply told which documents hallucinate already scores higher (0.82) than our forecaster. We therefore test timing directly. Inside a single hallucinating document, where the document label tells you nothing, the forecaster still ranks the tokens just before an onset above earlier faithful ones, at 0.687 AUROC. This within-document signal is the real *when*, and no prompt-level model has it.

For RQ2, the early warning is text-novelty drift, not language-model surprisal. A hazard head that sees only the 18 text features matches the full causal set; the language-model features alone trail it. The features that best tell an onset seen coming from an abrupt one are all novelty-window statistics. In short, the model drifts away from its source, producing increasingly out-of-context text, before it hallucinates. Two alternative explanations fail. The signal is not autocorrelation: a matched-capacity model that sees only the history of past onsets reaches 0.65, but the content features beat it by 0.13 AUROC at every horizon. Nor is the onset process self-exciting: a Hawkes excitation term adds nothing once we control for each document’s onset rate.

For RQ3, forecasting reaches what detection cannot. At a matched false-alarm budget and matched recall, the forecaster raises its alarm a median of 11 tokens *before* the onset, where the companion detector fires 15 tokens *after*. The sign of the delay flips, against a bound that forbids negative delay for a detector. The early warning is, however, domain-specific. The forecaster transfers to a short-form question-answering corpus only when refit in-domain, where it again reaches 0.740; it fails zero-shot, whereas detection partially transfers [12]. Timing is more particular to a domain than detection is.

Our contributions:

1. We formulate hallucination-onset forecasting as causal, token-level, black-box discrete-time survival, with a streaming-safe feature audit and a forward-only hazard head (Sections 3, 4).
2. We show the onset is forecastable (0.78 AUROC at horizon $k=3$, five seeds), and we separate the forecast into a between-document part and a genuine within-document timing signal that no prompt-level predictor can supply (Section 5.1).
3. We locate the early warning in text-novelty drift, not language-model surprisal, and rule out two alternatives, onset autocorrelation and self-excitation, with a matched-capacity null and a within-document permutation null (Sections 5.2, 5.3).
4. We show the forecaster reaches negative delay against the companion detector at matched false alarm (Section 5.4), and we characterize what precedes a forecastable onset, why an abort policy does not need the lead time, and why self-consistency does not help (Sections 5.5–5.7).

2 Related Work

Our task sits between three lines of work: token-level hallucination detection, which supplies the labels and features; pre-generation risk estimation, which shares the goal of warning early but works at the prompt level; and sequential change-point monitoring, which gives the streaming framing. We review each in turn and end with the baselines we compare against.

Token-level detection and its onset. Hallucination in generation is surveyed by Ji et al. [13]. The dominant families score a finished response: by sampling consistency [23], by calibrated uncertainty [14, 18, 8], or by probing internal state [3, 6]. All of them judge text that already exists, which is the property this paper sets out to relax. Token-level reference-free detection was framed as a benchmark task by Liu et al. [21], and corpora such as RAGTruth [26] now provide span-level labels at scale. Most detectors are scored as classifiers, by area under the ROC curve over all tokens; recurrent models over the log-probability stream have also been applied at the response level [34]. The one result closest to ours already points to the onset as special: Snel and Oh [37] show that the first token of a hallucination span is far more detectable than its continuation, an AUC near 0.8 against near 0.5. That is a change-point statement, and it is what makes forecasting plausible: the signal concentrates where the state switches. Where the switch comes from has also been studied. Akarlar [1] give causal evidence that a hallucinated state propagates forward through autoregressive conditioning, corrupting the continuation far more than a repair restores it. Alvarez and Baheri [2] locate the first error in a reasoning chain as an excursion in hidden-state geometry. The operational case for reacting during generation is made by Obeso et al. [27], who flag hallucinated entities in real time. None of these predicts the onset before it occurs, which is our question.

Pre-generation, prompt-level risk. A parallel line estimates hallucination risk before decoding, but at the prompt or response level and from model internals. Miftakhova and Zaytsev [24] estimate a per-prompt hallucination probability from internal representations, with a soft-target objective, before any token is produced. Roy et al. [31] find that activation probes carry a statistically significant signal already at position zero, useful for pre-generation flagging even where they do not beat output-based detectors on raw accuracy. Kogilathota et al. [17] predict hallucination risk in vision-language models in a single forward pass. All three are prompt- or response-level, white-box, and static: they estimate *whether* a generation is risky. We are complementary and orthogonal: token-level, online, black-box, and predicting *when and where* inside a generation the risk becomes

an onset. A prompt-level score chooses which generations to watch; our forecaster says when to intervene inside one.

Sequential monitoring, and the ladder this paper extends. Treating generation as a stream invites sequential change-point detection, whose false-alarm–delay trade-off and delay bounds are classical [28, 35, 22, 25, 19, 29]. That theory prices the delay *after* a change and says nothing about the interval before one, because it assumes the pre-change stream carries no information about when the change will come. Proposition 1 draws out what that assumption implies, and Section 5.4 measures how badly it fails for language-model generation. Xie [40] argues that such sequential tools should become standard for monitoring deployed language models, including shifts in hallucination rates across queries. We work one level down, inside a single generation. This paper is the third in a sequence built on the same 33-dimensional black-box feature stream. The first localizes hallucinated spans at the token level [12]. The second formulates onset *detection* as quickest change detection and proves a delay bound of about 1.3 tokens after onset [11]. Here we ask whether the onset can be *forecast* before it begins, which no prior work in this line does.

Baselines. Two comparisons anchor our claims. For the forecast itself, the natural floor is a pointwise (non-recurrent) hazard on the same features, which isolates the contribution of the temporal trajectory. A per-document oracle bounds the between-document signal a prompt-level predictor could reach. For the payoff, the natural comparison is the companion onset detector [11], run at a matched false-alarm budget, which lets us ask whether forecasting achieves the negative delay a detector cannot.

3 Problem Formulation

Onset as a failure event. A generation is a token sequence $x_1, \dots, x_n$ with a binary faithfulness label $s_t \in \{0, 1\}$ per token, $s_t = 1$ marking a hallucinated token. A hallucinated *span* is a maximal run of $s_t = 1$; its first token is an *onset*. Write $\mathcal{O}$ for the set of onset positions and $\theta = \min \mathcal{O}$ for the first one, the document’s change point ($\theta = \infty$ if the generation is faithful throughout). A token is in the *risk set* if it is still faithful, $s_t = 0$: these are the tokens at which an onset can still occur in the future. Tokens already inside a span are in the event state and are excluded, just as survival analysis removes subjects that have already failed.

Discrete-time hazard. Let $\phi_{\leq t}$ denote the causal feature history available up to and including token t (Section 4). For a horizon k we define the discrete-time hazard

$$h_t^{(k)} = \mathbb{P}(\exists o \in \mathcal{O} : t < o \leq t + k \mid \phi_{\leq t}, s_t = 0), \quad (1)$$

the probability that a span begins within the next k tokens, given everything observed so far and given that the current token is still faithful. This is a discrete-time hazard in the sense of Cox [7], Brown [4] and Kalbfleisch and Prentice [15], evaluated over the risk set; the multi-horizon form is the discrete-time analogue of the dynamic prediction of van Houwelingen [39]. We report the four horizons $k \in \{1, 3, 5, 10\}$: $k = 1$ is the immediate next-token onset probability, $k = 10$ a longer-range forecast. A predictor of (1) is a *forecaster* rather than a detector: at token t it commits to a statement about tokens it has not yet seen, and it is scored on whether the onset arrives. Casting the task as a hazard over a risk set is not only notation: conditioning on $s_t = 0$ makes the exclusion of in-span tokens principled rather than ad hoc, and expressing the forecaster as a per-token hazard puts it on the same time axis as the change-point detector it is later compared against (Section 5.4).

Streaming validity. Equation (1) conditions on $\phi_{\leq t}$ only. For the predictor to be usable during generation, every feature in $\phi_{\leq t}$ must be computable from the prefix $x_1, \dots, x_t$ alone, and the model reading it must not look ahead. We enforce both: a causal feature audit (Section 4) removes any component normalized by, or computed from, the full output, and the hazard head is a forward-only recurrent network. A forecaster that used bidirectional context or a length-normalized position would report an optimistic number it could never achieve online; we treat that configuration only as a leakage ceiling.

Evaluation. The label at token t for horizon k is $y_t^{(k)} = \mathbf{1}[\exists o : t < o \leq t + k]$, defined over risk-set tokens. We score a hazard predictor by the area under the ROC curve (AUROC) and the area under the precision–recall curve (PR-AUC) of $h_t^{(k)}$ against $y_t^{(k)}$ over the test risk set, per horizon. Pooled AUROC ranks all risk-set tokens together and therefore blends two distinct signals: whether a document hallucinates at all, and when within a hallucinating document an onset is imminent. We separate the two in Section 5.1, since only the second is a timing claim. When we compare a forecaster’s alarm to a detector’s, the common currency is the false-alarm rate on clean documents, one false alarm per $1/\alpha$ clean tokens (Section 5.4), matching the operating-point convention of the companion detector [11].

What a detector cannot do. A detector and a forecaster are both alarm rules: stopping times T on the prefix filtration $\mathcal{F}_t = \sigma(x_1, \dots, x_t)$. They differ in what the still-faithful prefix is assumed to carry. Sequential change detection treats it as uninformative about the change point: conditionally on $\theta = m$, the first $m - 1$ tokens follow the law of a stream in which the change never arrives [28, 35, 19]. That assumption is what makes the classical theory tractable, and it is not innocuous here. Under it an alarm before the onset is by construction a false alarm, so the false-alarm budget alone caps how often one can occur.

Proposition 1 (Early-warning ceiling). *Let T be an $(\mathcal{F}_t)$ -stopping time that fires on the clean stream with per-token probability at most α , so $\mathbb{P}_\infty(T \leq m) \leq \alpha m$ for every m . Suppose the pre-onset prefix is uninformative, meaning $\mathbb{P}(A \mid \theta = m) = \mathbb{P}_\infty(A)$ for every $A \in \mathcal{F}_{m-1}$. Then*

$$\mathbb{P}(T < \theta) \leq \alpha (\mathbb{E}[\theta] - 1). \quad (2)$$

Proof. Because T is a stopping time, $\{T < m\} = \{T \leq m - 1\} \in \mathcal{F}_{m-1}$, so the uninformative-prefix identity applies to it and gives $\mathbb{P}(T < m \mid \theta = m) = \mathbb{P}_\infty(T \leq m - 1) \leq \alpha(m - 1)$. Averaging over the law of θ ,

$$\mathbb{P}(T < \theta) = \sum_m \mathbb{P}(\theta = m) \mathbb{P}(T < m \mid \theta = m) \leq \alpha \sum_m \mathbb{P}(\theta = m) (m - 1),$$

which is (2). □

The bound is tight: a monitor that ignores the data entirely and alarms with probability α at each token attains it up to the union-bound slack. It is also conservative in the direction that matters here. We obtain $\mathbb{P}_\infty(T \leq m) \leq \alpha m$ from a union bound over the first m clean tokens, which over-counts, so a measured violation of (2) understates how far the premise fails.

Remark 1. Equation (2) constrains the pre-onset side of the delay distribution. Lorden’s bound constrains the post-onset side, $\mathbb{E}[(T - \theta)^+] \gtrsim \log(1/\alpha)/D$ for the Kullback–Leibler separation D between the two regimes [22]. Together they hold an uninformative-prefix monitor on the nonnegative side of the onset: it seldom fires early, and when it fires late it cannot be arbitrarily

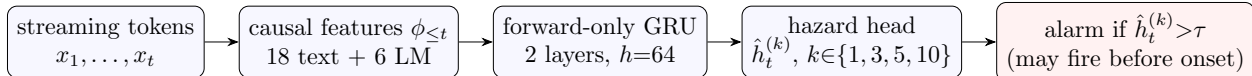


Figure 1: The forecasting pipeline. Everything to the left of the hazard head is causal: the hazard at token t depends only on the prefix $x_1, \dots, x_t$, so the predictor runs during generation. The alarm can precede the onset because the hazard is a forward-looking statistic, not an accumulation of past evidence.

fast. A negative median delay at a tight budget is therefore not evidence of a sharper detector. It is evidence that the premise fails, that the still-faithful prefix carries information about θ , which is the same statement as the hazard (1) varying over the risk set rather than sitting at the marginal onset rate. The gap between (2) and the realized rate of pre-onset alarms is then a measurement of how far it fails, and Section 5.4 reports it for both a forecaster and a detector.

4 Method

Figure 1 shows the pipeline. A streaming decoder emits tokens; from the prefix up to token t we compute causal features, read them with a forward-only recurrent network, and emit a per-horizon hazard. An alarm fires when the hazard crosses a threshold, which can happen before the onset.

Features and the causal audit. The feature stream is the 33-dimensional black-box representation of the temporal detector [12]: 20 text features (in-context / novel n-gram overlap with the source, novelty windows of widths 5/10/20, entity and repetition flags), 7 sentence-level natural-language-inference features from a DeBERTaV3 entailment model [10], and 6 language-model features (token log-probability, entropy, and derived quantities). None requires access to the generator’s weights. For forecasting, however, three groups are not causal and are removed. The position feature is $i/(n-1)$, normalized by the *total* output length, unknown mid-stream. The sentence-position feature is normalized by the total number of sentences, likewise unknown. The seven NLI features are computed per sentence and projected onto every token of that sentence, including tokens emitted before the sentence is complete, so a token’s NLI value depends on words not yet generated. Dropping these leaves 24 streaming-safe features: 18 text and 6 language-model. We report the full 33 only as a leakage ceiling.

Hazard head. Reading a discrete-time hazard off a neural encoder is standard in survival modelling [16, 20, 30]; what is new here is the event being modelled and the constraint that every input be available mid-stream. The predictor is the companion’s exact recurrent labeler [11, 5]: a two-layer forward-only GRU with hidden width 64 and dropout 0.1 (47K parameters). We replace its classification head with a hazard head that emits one logit per horizon, $\hat{h}_t^{(k)} = \sigma(g_k(r_t))$. Here r_t is the recurrent state after reading $\phi_{\leq t}$, and g is a two-layer network (Linear–GELU–Linear) with four outputs. Because the GRU is forward-only, r_t depends on nothing after t , so the head is causal by construction. We train with a masked multi-horizon binary cross-entropy over the risk set,

$$\mathcal{L} = \frac{1}{\sum_t \mathbf{1}[s_t = 0]} \sum_{t: s_t = 0} \sum_k w_k \text{BCE}(\hat{h}_t^{(k)}, y_t^{(k)}),$$

with a per-horizon positive weight w_k equal to the negative/positive ratio in the training risk set (hazards at longer horizons are less rare). Tokens inside a span are masked out of the loss.

Optimization follows the companion detector’s recipe [11]: AdamW at learning rate 10^{-3} , weight decay 10^{-4} , with early stopping on validation $k=3$ AUROC, in under a minute per seed on one RTX 4090. A pointwise logistic hazard on the same features and labels is the linear floor.

Data. We use RAGTruth [26] with its standard split, the corpus the temporal detector was built on. Its generations span three tasks, question answering, summarization, and data-to-text, each with token-level faithfulness labels. The training risk set holds 1.87M faithful tokens, the test risk set 327K over 2700 documents. For the cross-domain study we use the English subset of PsiloQA [32], a short-form question-answering corpus with span-level hallucination labels. Its language-model medians and feature standardization come from its own training split, cached separately so that a prompt normalized under one corpus is never scored under another’s.

Two null models. Onset positions cluster within documents, so a forecaster could score above chance by learning that clustering rather than any content precursor. We control for this in two ways. The *matched-capacity* null feeds the same GRU trunk only causal label and clock features (token position, tokens since the last onset, number of prior onsets, running hallucination rate). It can therefore exploit all the temporal structure of the onset process itself, but sees no content. If the content-feature hazard beats it, the extra signal is not autocorrelation. The *self-excitation* null fits a discrete-time logistic point process $\lambda_t = \sigma(w^\top \phi_t + \sum_j a_j e_t(\beta_j) + \gamma f_d + b)$. Here $e_t(\beta) = \sum_{o < t} e^{-\beta(t-o)}$ is an exponentially decaying sum over past onsets (a discrete Hawkes term, 9) at fixed decay rates $\beta \in \{1, \frac{1}{3}, \frac{1}{10}\}$, and f_d is a per-document frailty. We test the excitation coefficients against a within-document permutation null that shuffles onset positions while keeping their count, isolating temporal contagion from per-document heterogeneity.

Lead-time calibration. To compare a forecaster’s warning to a detector’s, both must run at the same false-alarm budget. The forecaster’s alarm is the first token whose hazard score $\hat{h}_t^{(k)}$ crosses a threshold; the detector is the companion’s ForwardGRU-CUSUM, whose statistic accumulates log-odds and crosses a threshold of its own [11, 22, 28]. We split the test documents into a calibration and an evaluation half, stratified by clean versus hallucinated. Thresholds and the CUSUM reference value are fit on calibration only, then frozen and measured on evaluation. We report the realized evaluation false-alarm rate: the two fall within about 20% of the target and of each other (for example 0.0009 vs 0.0011 at $\alpha=10^{-3}$), so the budgets are matched up to discretization. The signed delay of an alarm at token a relative to onset θ is $d = a - \theta$; $d < 0$ means the alarm preceded the onset. We report recall alongside delay, so an early median cannot be a collapsed-recall artifact. We compare the two detectors on the onsets caught by both, with a one-sided paired Wilcoxon test and a directional McNemar test.

5 Experiments

The experiments take up the three research questions of Section 1 in order: whether onset is forecastable and on what horizon (RQ1), where the early warning lives (RQ2), and whether a forecaster reaches a lead time no detector can (RQ3). We close with the ablations that rule out trivial explanations of the signal.

	$k=1$	$k=3$	$k=5$	$k=10$
<i>Recurrent hazard (5 seeds, mean$\pm$std)</i>				
causal-24	0.805 $\pm$.003	0.777 $\pm$.002	0.770 $\pm$.002	0.761 $\pm$.002
text-18 (causal)	0.797 $\pm$.003	0.776 $\pm$.002	0.770 $\pm$.002	0.761 $\pm$.002
lm-6	0.757 $\pm$.002	0.747 $\pm$.002	0.743 $\pm$.002	0.739 $\pm$.002
full-33 (ceiling)	0.813 $\pm$.002	0.789 $\pm$.002	0.782 $\pm$.002	0.775 $\pm$.002
<i>Baselines</i>				
logistic hazard (causal-24)	0.683	0.657	0.647	0.639
position-only	0.495	0.486	0.479	0.461
oracle doc-constant	0.814	0.816	0.819	0.825
<i>Within-document (timing only)</i>				
causal-24 forecaster	0.740	0.687	0.669	0.649
position-only	0.553	0.545	0.537	0.516
any doc-constant	0.500	0.500	0.500	0.500

Table 1: Hazard AUROC by horizon on the RAGTruth test risk set. The recurrent forecaster reaches 0.78 at horizon $k=3$ (onset within three tokens); the causal features nearly match the leaking ceiling. Pooled AUROC has a between-document component (an oracle that knows only the document label scores 0.82), so we also report the within-document AUROC, where any per-document predictor (including that oracle) is 0.5 by construction.

5.1 Onset is forecastable, and the forecast has a timing component

Table 1 gives the hazard AUROC per horizon. A pointwise logistic hazard on the 24 causal features already clears chance at every horizon (0.657 at $k=3$), so onset is forecastable from causal features alone. The recurrent hazard head lifts this by twelve points to 0.777 ± 0.002 at $k=3$ (mean and standard deviation over five seeds), and stays above 0.76 out to $k=10$, ten tokens ahead. The gain over the logistic floor is not from the features but from reading their *trajectory*: the same features scored pointwise give 0.657, scored recurrently 0.777. The causal features lose almost nothing to the full, future-leaking set (0.777 vs 0.789), so the anticipatory signal is available online.

Between-document signal versus within-document timing. A pooled AUROC answers two questions at once: does this document hallucinate, and when inside it does the onset come. Most positive tokens sit in hallucinating documents. A predictor that only knew which documents those are would already separate much of the risk set. To measure that part, we give an oracle the true document label, “does this document hallucinate,” which uses the future. Spread as a constant over the document’s tokens, the oracle scores 0.816 at $k=3$, above the causal forecaster’s 0.777; a position-only score is at chance (0.486). Part of the pooled number, then, is just knowing which document is risky, and a prompt-level predictor knows that too. The timing claim needs a cleaner test. We restrict to one hallucinating document at a time ($n=934$ documents), where the document label is constant and carries no information. There the forecaster ranks the tokens just before an onset above earlier faithful ones at 0.687 AUROC (bootstrap 95% CI [0.672, 0.702]; 0.687 ± 0.003 across five seeds), against 0.545 for position. The online forecaster therefore does two things: it recovers most of the oracle’s between-document signal without seeing the future (0.777 vs 0.816), and it adds a within-document timing signal (0.687 vs 0.5) that no prompt-level model can give. This answers RQ1.

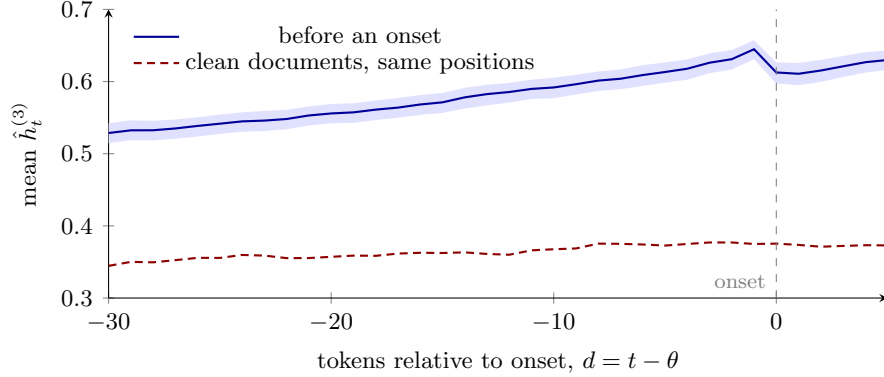


Figure 2: The hazard rises before the onset. Mean $\hat{h}_t^{(3)}$ over the 1396 held-out onsets with at least 10 tokens of runway, aligned at $d=0$, with a document-clustered bootstrap band. It climbs +0.116 over the last 30 faithful tokens. The control curve reads the same absolute positions in clean documents, where it is flat (+0.030), so the rise is a precursor rather than a position effect. Tokens at $d \geq 0$ sit inside the span and outside the training risk set.

5.2 The precursor is text-novelty drift, not language-model surprisal

Figure 2 shows the shape of the warning. Aligning every held-out onset at $d=0$, the mean hazard climbs monotonically from 0.529 thirty tokens out to 0.645 at the last faithful token, a rise of +0.116. The climb alone would not settle much, since a hazard that simply grew with position would produce the same picture, so the figure carries a control: the same absolute positions read from documents that never hallucinate. That curve is flat, 0.345 to 0.375, and the bootstrap band around the onset-aligned curve stays clear of it across the whole window. The rise is a precursor, not a position effect.

Figure 3 shows the features underneath it. The three novelty windows drift up together before the onset, the width-20 window by +0.126 standardized units, and then rise a further +0.195 in the five tokens after it. Most of the novelty signal is post-onset. That asymmetry is worth stating plainly, because it is the quantitative reason forecasting is the harder problem: a detector waits for the jump, while a forecaster has only the leading edge of the drift to work with. It also explains why a within-document forecast at 0.687 sits well below what a detector reaches on the very same feature stream, without either number being anomalous.

The feature ablation in Table 1 locates the early warning. A hazard head that sees only the 18 causal text features matches the full causal set (0.776 vs 0.777 at $k=3$): the language-model block adds nothing at the margin. The language-model features alone still forecast at 0.747, well above chance, and well above their logistic value of 0.599. The recurrent model therefore does read an early warning from the language-model trajectory, but text drift already contains it. The point is not that language-model uncertainty is useless for onset. It is that novelty drift already carries what it would add. This separates forecasting from detection. In detection, language-model entropy spikes *at* the onset token [12]. In forecasting, the early signal is the model drifting away from its source, producing more and more out-of-context text, not a local uncertainty spike.

5.3 Not autocorrelation, not self-excitation

Onsets cluster within documents, so the forecast could in principle be autocorrelation of the onset process, not a content warning. The latent faithful/hallucinated state is close to a first-order chain

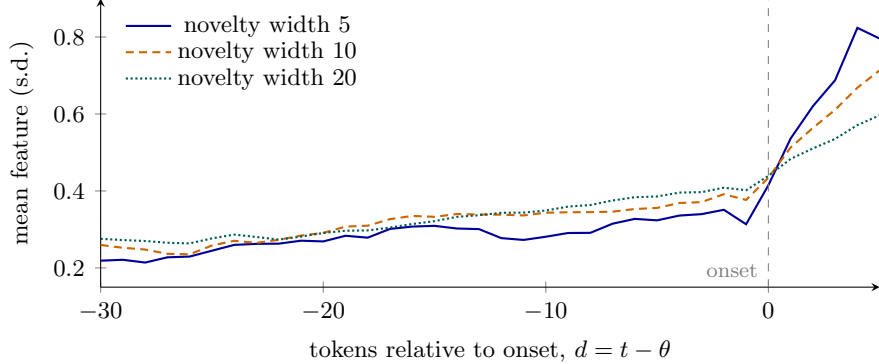


Figure 3: The novelty windows on the same onset alignment, in standardized units. They drift up before the onset (the width-20 window by $+0.126$ over the last 30 faithful tokens) and then rise $+0.195$ more in the five tokens after it. The pre-onset climb is what a forecaster can use; the post-onset jump, several times larger, is what a detector sees. The asymmetry is why forecasting scores below detection on the same feature stream.

[11]; we estimate its second eigenvalue at 0.90, which limits how far the state history alone can forecast. The matched-capacity null makes this concrete. We give the same recurrent trunk only the causal history of onset labels and clocks. It reaches 0.65, so onset clustering is genuinely forecastable, but the content-feature hazard beats it by 0.13 AUROC at every horizon (paired document-clustered bootstrap CI $[+0.117, +0.145]$ at $k=3$). The early warning is content, not just the shadow of past onsets.

Nor is the onset process self-exciting. We fit a discrete-time logistic point process with a Hawkes excitation term over past onsets. The excitation improves out-of-sample intensity over a covariate-plus-frailty model by only $+0.011$. However, a within-document permutation null, which shuffles onset positions while keeping their count, improves out-of-sample intensity by $+0.026$ on average—*more* than the real excitation term itself, so the one-sided test finds no evidence of contagion ($p=1.000$). The apparent excitation is per-document variation in onset rate, not temporal contagion. Within-span persistence does not carry over to the onset process, which is why we model the hazard from covariates rather than from an onset self-history. Together with Section 5.2, this answers RQ2: the early warning is text-novelty drift, and it is neither autocorrelation nor self-excitation.

5.4 Negative-delay forecasting against the onset detector

The forecaster’s payoff is a warning before the event, which a detector cannot give. We compare its alarm to the companion ForwardGRU-CUSUM at a matched false-alarm budget on the held-out evaluation half (Table 3). At a tight budget of one false alarm per 1000 clean tokens, with recall matched near 0.46, the forecaster’s median signed delay is -11 tokens: it warns eleven tokens *before* onset, where the detector fires fifteen tokens *after*. The sign of the delay flips (Figure 4). On the onsets caught by both, the forecaster is earlier by a one-sided paired Wilcoxon test ($p=2.6 \times 10^{-19}$), and a directional McNemar test counts 62 onsets the forecaster warns early against 2 for the detector. (At a much looser budget of $\alpha=10^{-2}$ the detector’s threshold is low enough that it alarms almost immediately, before most onsets, so that regime is false-alarm-dominated rather than anticipatory; the meaningful comparison is at $\alpha=10^{-3}$.)

This realizes the negative delay the change-point bound structurally forbids: a detector reacts

quantity	value
evaluation hallucinating documents	472
pre-onset runway $\theta - 1$: mean / median / p90	62.4 / 51 / 119
document length: mean / median	147.6 / 139
forecaster realized α	9×10^{-4}
ceiling $\alpha(\mathbb{E}[\theta] - 1)$	0.056
measured $\mathbb{P}(T < \theta)$	0.36 (6.4 $\times$)
detector realized α	1.1×10^{-3}
ceiling $\alpha(\mathbb{E}[\theta] - 1)$	0.069
measured $\mathbb{P}(T < \theta)$	0.14 (2.0 $\times$)

Table 2: The early-warning ceiling of Proposition 1 against the measured rate of pre-onset alarms, horizon $k=3$ on the evaluation half. $\mathbb{P}(T < \theta)$ is recall times the fraction of caught onsets whose alarm precedes the onset. Each procedure is held to its own realized clean-stream false-alarm rate rather than the nominal budget.

statistic	recall	median delay	% caught before onset
forecaster	0.46	-11	0.78
detector	0.47	+15	0.29

Table 3: Signed alarm delay relative to onset (< 0 = before onset) at the matched false-alarm budget $\alpha = 10^{-3}$, forecaster vs the companion onset detector, on the onsets caught by both. Recall is matched (0.46 vs 0.47). “% caught before onset” is the fraction of the caught onsets whose alarm precedes the onset. The forecaster warns before onset while the detector fires after; paired Wilcoxon $p = 2.6 \times 10^{-19}$, directional McNemar 62 vs 2.

to the onset, so its delay is bounded below by a positive constant [22, 11]; a forecaster commits before the onset, so its delay can be negative.

Proposition 1 turns the sign flip into a quantity. On the evaluation half the pre-onset runway averages $\mathbb{E}[\theta] - 1 = 62.4$ tokens (median 51, Table 2), so a monitor whose still-faithful prefix were uninformative could warn ahead of at most $\alpha(\mathbb{E}[\theta] - 1) = 5.6\%$ of onsets at the forecaster’s realized budget of $\alpha = 9 \times 10^{-4}$. The forecaster warns ahead of 36% of them, exceeding that ceiling 6.4 $\times$. Applying the same accounting to the detector is instructive: at its own realized $\alpha = 1.1 \times 10^{-3}$ its ceiling is 6.9% and it warns ahead of 14%, or 2.0 $\times$. Both violate the bound, so the uninformative-prefix premise is not merely conservative for language-model generation, it is false, and a statistic built from a detection posterior already reads some pre-onset drift. What separates the two procedures is how much: the forecaster, which is trained to read it, recovers roughly three times as much. The margin by which the premise fails is, in this sense, a measurement of the early warning itself.

None of this hinges on the choice of detector. Against a bidirectional BiGRU detector (non-causal, and so strictly stronger than a streaming one) and against a linear detector, both still fire after onset at $\alpha=10^{-3}$ (median +18 and +42) while the forecaster fires before it (-11); the forecaster is earlier in every case (Wilcoxon $p \leq 5 \times 10^{-12}$). This answers the first half of RQ3: forecasting reaches a negative delay no detector can. The second half, whether it transfers, is taken up in Section 5.7.

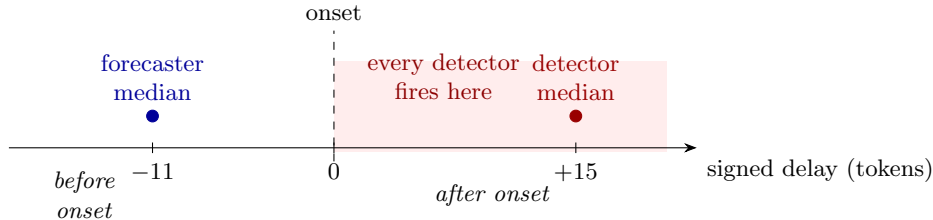


Figure 4: Operating point at a matched false-alarm budget ($\alpha=10^{-3}$) and matched recall. The forecaster’s median alarm falls 11 tokens before the onset; the companion detector’s falls 15 tokens after, in the post-onset region where a change detector’s delay is bounded below by a positive constant. Positions are schematic; the marked delays are the measured medians (Table 3).

feature	block	AUROC
width-10 novelty window	text	0.797
width-5 novelty window	text	0.788
width-20 novelty window	text	0.783
bigram overlap	text	0.246
running in-context ratio	text	0.248
running novel-token ratio	text	0.752
consecutive-novel counter	text	0.700
language-model signal flag	LM	0.314

Table 4: Single-feature separation of anticipated ($n=473$) from surprise ($n=473$) onsets, read at the last token before the onset and ordered by $|\text{AUROC} - 0.5|$. Values below 0.5 separate in the opposite direction: an anticipated onset carries *less* source overlap.

5.5 What precedes a forecastable onset

Not every onset is seen coming. For each held-out onset we read the hazard the forecaster gave at the last token before it. We call an onset *anticipated* if that hazard is in the top tercile, and *surprise* if it is in the bottom tercile. Of 1430 onsets, only 13 sit at the very first output token, with no runway. Anticipated onsets arrive earlier in the output (mean pre-onset hazard 0.68 early versus 0.57 late) and begin shorter spans (0.69 for spans of length ≤ 6 versus 0.58 for longer); late and long-span onsets are more abrupt. We then ask which pre-onset features separate the two groups. They are all text-novelty statistics: the width-10 novelty window separates them at 0.797 AUROC, width-5 at 0.788, width-20 at 0.783, and the running novel-token ratio at 0.752 (Table 4). Seven of the top eight are text features; the one language-model entry is a signal-availability flag, not surprisal.

A forecastable onset is one the model drifts into, with novelty building before the break. A surprise onset stays grounded in the source until the last faithful token. This is a sensitivity analysis of the forecaster, not independent evidence: the split is defined by the forecaster’s own hazard, so seeing novelty features separate it echoes Section 5.2 rather than confirming it afresh, and the top features are selected from the causal set without a multiple-comparison correction. What it adds is a concrete picture of *which* onsets the model sees coming.

stratum	documents	pooled	within-document
<i>Task type</i>			
data-to-text	900	0.790	0.741
question answering	900	0.732	0.649
summarization	900	0.626	0.564
<i>Generator</i>			
llama-2-13b-chat	450	0.797	0.725
mistral-7B-instruct	450	0.728	0.683
llama-2-7b-chat	450	0.719	0.680
gpt-3.5-turbo	450	0.663	0.673
llama-2-70b-chat	450	0.752	0.669
gpt-4	450	0.732	0.655

Table 5: Hazard AUROC at $k=3$ by RAGTruth stratum. Within-document scores average per-document AUROC over the documents that contain an onset, so a stratum’s within-document figure rests on its hallucinating documents only. The task spread is 0.177; the generator spread is 0.070.

5.6 Where the forecast holds: task, generator, calibration

RAGTruth spans three task types and six generators, and a precursor that existed for only one of them would be a weaker finding than one that held across them. Table 5 breaks the forecast down both ways, and the two answers differ.

Across generators the signal is stable. Within-document AUROC spans 0.655 to 0.725 over six models whose parameter counts differ by an order of magnitude, with GPT-4 and the Llama family in the same band. Whatever the forecaster is reading, it is not one model’s characteristic failure mode. Two of the six, GPT-4 and GPT-3.5, hallucinate in only 42 and 46 test documents, so their within-document figures rest on few documents and we report them for coverage rather than as precise estimates.

Across tasks it is not stable at all. Within-document AUROC is 0.741 on data-to-text, 0.649 on question answering and 0.564 on summarization, a spread of 0.177 against the 0.070 seen across generators. The headline 0.687 is an average that data-to-text dominates, since it supplies 578 of the 934 hallucinating documents, and on summarization the forecast is close to uninformative. The gradient is not arbitrary. It tracks how tightly an output is bound to its source, which is exactly what a novelty-based precursor needs: a data-to-text generation that starts inventing departs from a structured record immediately, whereas a summary can drift stylistically for many tokens before anything it says becomes unfaithful. Read that way, the zero-shot failure on PsiloQA in Section 5.7 is not a surprise arriving from outside the corpus but the continuation of a trend visible inside it, short-form question answering being already the weaker of the two RAGTruth tasks that resemble it.

The hazard ranks well and is not calibrated. The alarm thresholds $\hat{h}^{(3)}$, so it is worth asking what that number means. Over the 327K test risk-set tokens the empirical onset rate rises monotonically across all ten deciles of the score, from 0.002 in the lowest to 0.057 in the highest, a 28-fold spread against a base rate of 0.013: as a ranking it is sound. As a probability it is not. Mean predictions run from 0.11 to 0.79 over the same deciles, giving an expected calibration error of 0.387 and a Brier score of 0.196. This is by construction rather than by accident. The loss weights positives by the negative/positive ratio, which is what makes a 1.3%-positive problem learnable at all, and it buys that ranking at the cost of any probabilistic reading of the output. Everything we

	$k=1$	$k=3$	$k=5$	$k=10$
RAGTruth in-domain	0.810	0.783	0.775	0.767
PsiloQA zero-shot	0.434	0.383	0.349	0.319
PsiloQA refit in-domain	0.789	0.740	0.741	0.756
PsiloQA onset base rate	5.1%	14.3%	22.3%	36.1%

Table 6: Hazard AUROC across corpora; the PsiloQA risk set holds 37,591 tokens. Zero-shot transfer inverts and falls further with horizon, while the same architecture refit on PsiloQA matches its RAGTruth numbers, so the failure is one of transfer rather than of signal.

do with $\hat{h}$ uses it as a ranking, through AUROC or through a threshold fitted on held-out data, so no result here depends on the level; we call it a hazard score rather than a posterior for that reason. A deployment that needed the level, say to weigh an intervention’s cost against its expected benefit, would have to recalibrate on held-out data first.

5.7 Scope: transfer, mitigation, and self-consistency

Forecasting is domain-specific. Transferred zero-shot from RAGTruth to the short-form PsiloQA corpus, the forecaster inverts, giving AUROC 0.434/0.383/0.349/0.319, below chance and decreasing with horizon. This is neither a normalization artifact nor an absence of signal. It is not normalization: re-standardizing PsiloQA with its *own* feature statistics and language-model medians, keeping the RAGTruth weights, leaves it at 0.408 at $k=3$, still below chance. It is not signal-absence: refit in-domain on PsiloQA it reaches 0.789/0.740/0.741/0.756, comparable to its RAGTruth numbers (Table 6). What reverses is the mapping itself. We measure it as the point-biserial correlation of each causal feature with the imminent-onset label $y^{(3)}$ over the risk set, computed on each corpus under its *own* language-model medians and standardization, so that a reversal cannot be a normalization artifact; a feature counts as flipped when the two correlations carry opposite signs with $|r| > 0.02$ in both. Under that criterion 10 of the 18 causal text features flip, against 1 of the 6 language-model features (Table 7). Three properties make the reversal look systematic rather than incidental. The flips are exactly the novelty family, all three windows, the running novel and in-context ratios, the bigram and trigram overlaps and the consecutive-novel counter, which is the group that carries the forecast in RAGTruth. They flip in complementary pairs, the running novel ratio moving $+0.047 \rightarrow -0.163$ while its complement the running in-context ratio moves $-0.047 \rightarrow +0.163$, so the two corpora place the same variable on opposite sides of the onset. The reversal also amplifies: features at $|r| \approx 0.05$ in RAGTruth reach $|r| \approx 0.15$ in PsiloQA, three times as strong in the wrong direction, which is why a zero-shot forecaster does worse than chance instead of merely losing the signal. Mean correlations move the same way, from $+0.005$ to -0.026 over the text block, while the language-model block stays near zero in both (-0.007 and $+0.004$).

Rising novelty drift precedes an onset in long-form retrieval-augmented generation, but the same statistics point the other way in short-form question answering, so a RAGTruth-trained forecaster reads the PsiloQA signal backwards. Onset is forecastable in both corpora; the *direction* is domain-specific. Detection is less brittle: the temporal detector partially transfers across the same corpora (0.744) [12], so the timing signal is more domain-specific than the detection signal, forecasting and detection dissociating once again.

causal text feature	r RAGTruth	r PsiloQA
width-20 novelty window	+0.049	−0.163
running in-context ratio	−0.047	+0.163
running novel ratio	+0.047	−0.163
width-10 novelty window	+0.046	−0.152
width-5 novelty window	+0.040	−0.139
bigram overlap	−0.035	+0.081
trigram overlap	−0.028	+0.064
in-context token	−0.021	+0.086
novel token	+0.021	−0.086
consecutive-novel counter	+0.021	−0.084

Table 7: The causal text features whose correlation with imminent onset ($y^{(3)}$ over the risk set) reverses between the corpora, largest $|r|$ in RAGTruth first. Each corpus is standardized with its own language-model medians, so the reversal is not a normalization artifact. A feature counts as flipped when the two correlations carry opposite signs with $|r| > 0.02$ in both.

α	policy	averted	cut	documents stopped
10^{-2}	forecaster	100.0%	100.0%	1351
	detector	87.3%	57.7%	1021
	random	59.1%	50.3%	1351
10^{-3}	forecaster	30.0%	16.2%	261
	detector	34.2%	13.3%	323
	random	17.2%	12.2%	261

Table 8: Circuit-breaker simulation at horizon $k=3$ over the evaluation half (879 clean and 472 hallucinating documents). “Averted” is the share of hallucinated tokens cut off, “cut” the share of faithful tokens lost. At the loose budget both policies stop nearly every document and the comparison degenerates.

An abort policy does not need lead time. We have no live decoder here, so the only mitigation we can simulate offline is a circuit-breaker: halt the continuation at the first alarm. At a matched false-alarm budget of 10^{-3} , halting on the forecaster averts 30.0% of hallucinated tokens at a cost of cutting 16.2% of faithful tokens; halting on the detector averts 34.2% at a cost of 13.3%; a random-stop policy matched to the forecaster’s stop set averts only 17.2% (Table 8).

The forecaster’s alarm timing clearly beats random, but for a pure *abort* the post-onset detector is more token-efficient: it aborts the same span tail without cutting the faithful tokens the forecaster sacrifices before onset. Lead time helps only actions that must fire before the onset (regenerating or constraining the span rather than discarding the rest of the generation), which require model access and are out of scope for an offline study. We report this to scope the mitigation claim rather than to inflate it.

Self-consistency does not help. A natural richer feature is black-box self-consistency: resample the output $K=5$ times and measure disagreement [23]. This is an offline, non-streaming measurement. On the subset for which we computed it (a single open-weight generator), adding the two consistency features to the forecaster changes AUROC by at most 0.004 at every horizon over five seeds, within one standard deviation (Table 9).

The result is a clean null. The text-novelty features already carry what a consistency resample

k	causal-24	+ consistency	Δ
1	0.684 $\pm$.007	0.684 $\pm$.007	-0.000
3	0.654 $\pm$.005	0.657 $\pm$.006	+0.003
5	0.642 $\pm$.004	0.646 $\pm$.005	+0.004
10	0.626 $\pm$.005	0.630 $\pm$.003	+0.004

Table 9: Adding two black-box self-consistency features to the forecaster, mean and standard deviation over five seeds on the 2515 documents for which resamples were available. Every change sits inside one standard deviation. Absolute values are lower than in Table 1 because this subset is smaller and its split is internal.

would report. Self-consistency is a control, not a missing ingredient.

6 Discussion

The three papers in this series measure the same event on a shrinking clock. Detection asks whether a token is hallucinated. The change-point analysis asks how many tokens pass before a monitor reacts, and proves that number has a positive floor. Forecasting moves the clock past zero, warning before the first false token, and Proposition 1 says why that is a difference in kind rather than in degree: it needs the pre-onset stream to carry information that change detection assumes away. Early-warning statistics ahead of a critical transition are a familiar object outside NLP [33]; what supplies one here is specific, and it is the finding we think most interesting. The shift is not cosmetic: a monitor that only reacts can log or redact after the fact, while a monitor that anticipates can act during generation. The mechanism behind it is specific, and we think it is the most interesting finding here. The model does not enter a hallucination abruptly from a faithful state. It drifts toward one, with novelty relative to the source building over several tokens before the break. That drift, not a local uncertainty spike, is what the forecaster reads. Language-model surprisal marks the onset; text drift precedes it.

This is also why forecasting and detection keep dissociating. The two rely on different signals, drift versus surprisal. They behave differently under every test we ran: the language-model block is decisive for neither the forecast margin nor its transfer, and the timing signal is more domain-specific than the detection signal. It is tempting to fold forecasting into detection as “early detection,” but they are different measurements of different quantities, and merging them would hide the drift mechanism that makes the forecast work.

We are deliberately cautious about mitigation. It would be easy to present the negative-delay result as a drop-in safety fix. Our own abort simulation argues against that. For a policy that simply halts generation, a post-onset detector is as good or better, because aborting the tail of a span does not require having seen it coming. Lead time is valuable only for actions that must commit before the onset: regenerating the span, constraining the next tokens, or routing to a more careful decoder. Testing those needs a live generator, which we do not have here. The honest claim is therefore narrow. The early warning exists, it is causal and black-box, and it reaches negative delay; turning it into a lower hallucination rate is future work that needs model access.

Finally, the between- versus within-document split is worth carrying into how this line is evaluated. A pooled token-level AUROC rewards knowing which generation is risky, which is exactly what prompt-level pre-generation predictors already estimate [24, 31]. Our contribution is orthogonal to that: when, inside a flagged generation, to expect the onset. Only the within-document number measures it. The two axes compose. A prompt-level score picks the generations to watch; a

token-level forecaster says when to intervene inside them.

7 Conclusion

We cast hallucination-onset forecasting as causal, token-level, black-box discrete-time survival, and showed that the onset can be predicted before it begins: a forward-only recurrent hazard reaches 0.777 AUROC three tokens ahead, over five seeds, using only streaming-safe features. The forecast carries a genuine within-document timing signal (0.687 AUROC) that no prompt-level predictor can supply, it lives in text-novelty drift rather than language-model surprisal, and it survives a matched-capacity autocorrelation null and a Hawkes self-excitation null. At a matched false-alarm budget the forecaster warns eleven tokens before onset where the companion detector fires fifteen after, ahead of 6.4 times as many onsets as any monitor could whose still-faithful prefix carried no information about the onset, a ceiling we prove. The warning is graded by task, near chance on summarization, and it does not transfer zero-shot, and lead time pays off only for interventions that act before the onset. A detector can only react once a hallucination has begun; the forecaster warns while the text is still faithful, because the model drifts toward a hallucination over several tokens rather than entering it abruptly.

Limitations

The study rests on two corpora, RAGTruth and the English subset of PsiloQA, both in English. Whether the drift precursor holds in other languages or generation regimes is open, and the domain-specific transfer of Section 5.7 is a reason for caution. The headline within-document number is also a task average rather than a uniform property: it is 0.741 on data-to-text and only 0.564 on summarization (Section 5.6), so on the least source-bound task the forecast is close to uninformative, and the method should not be read as working equally everywhere. The hazard is a ranking statistic rather than a calibrated probability (expected calibration error 0.387), which is adequate for the thresholded alarm studied here and inadequate for any use that needs the level. The feature stream and the architecture are inherited from prior work, where they were chosen by ablation [12, 11] rather than re-researched here. A different feature set could raise the numbers or shift which signals dominate. The text-versus-language-model dissociation, though, is consistent across our logistic and recurrent analyses. The mitigation result is an offline abort simulation, not a live intervention. We can show that the early warning exists and reaches negative delay, but not that acting on it lowers the hallucination rate. That needs a decoder in the loop. The self-consistency null was measured on a single open-weight generator, so it bounds the effect for that setting rather than universally. Finally, the within-document timing metric is defined only on documents with at least one onset and one faithful token; the 13 of 1430 onsets at the first output token have no runway and are excluded there.

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